

Door Control System - High Level Design

Prototype Evaluation Document - Synthetic Data

Field	Value
Document ID	HLD-DOOR-001
Version	1.0
System	Door Control System
Classification	Synthetic / Evaluation Only

1. Purpose

This High Level Design describes the software architecture of the Door Control System. The document defines software components, ports, interfaces, signals, dependencies and high-level functional flow for prototype evaluation.

2. Software Components

Component	Responsibility
DoorSwitch	Provides the driver door-lock request.
DoorController	Processes door-lock requests.
DoorLockActuator	Controls physical door locking.

3. Interfaces

Interface	Provider	Consumer
DoorRequestInterface	DoorSwitch	DoorController
DoorLockInterface	DoorController	DoorLockActuator

4. Port Definitions

Component	Port	Port Type	Interface
DoorSwitch	DoorRequestOut	P-Port	DoorRequestInterface
DoorController	DoorRequestIn	R-Port	DoorRequestInterface
DoorLockActuator	DoorLockCommandIn	R-Port	DoorLockInterface

5. Signal Definitions

Signal	Interface	Unit
DoorLockRequest	DoorRequestInterface	boolean
DoorLockCommand	DoorLockInterface	boolean

6. Component Dependencies

Dependent Component	Required Component
DoorController	DoorSwitch
DoorLockActuator	DoorController

7. Functional Flow

DoorSwitch -> DoorController -> DoorLockActuator

8. Revision Information

Current document version: 1.0. Document identifier: HLD-DOOR-001.